

MCS-212

The Discrete Mathematics Assignment Number

Q-1 Use Mathematical induction to prove that:

$$1^2 + 2^2 + \dots + m^2 = m(m+1)(2m+1)/6$$

Ams- To solve the problem, we'll use mathematical induction to prove that

$$1^2 + 2^2 + \dots + m^2 = \frac{m(m+1)(2m+1)}{6}$$

Base Case ($m=1$):

$$\text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1(1+1)(2(1)+1)}{6}$$

$$= \frac{6}{6} = 1$$

Since $\text{LHS} = \text{RHS}$, the base case is true.

Inductive Step:-

Assume the formula is true for

$$m = k$$

$$P(k): 1^2 + 2^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

we need to prove it for

$$m = k+1$$

Q-2 How many different permutations are possible of the letters, taken all at a time, of the word: ASSESSES?

=> The word "ASSESSES" has a total of 8 letters. To find the number of different permutations of the letters, taken all at a time, we can use the formula for permutations of a set of distinct elements, which is $n!$ (n factorial).

For the word "ASSESSES".

- There are 4 'S', so we need to account for the repetition.

o The total number of distinct letters is $8 - 4 = 4$ (A, E, S, E).

The number of permutation is given by $4!$ (4 factorial) since the 'S' are indistinguishable.

Number of permutations =

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

So, there are 24 different permutations of the letters in the word "ASSESES" when taken all at a time.

$$\frac{(k+1)(k+2)(2k+3)}{6} + (k+1)^2$$

$$\Rightarrow \frac{(k+1)(k+2)(2k+3)}{6} + \frac{(k+1)^2 \cdot 6}{6}$$

$$\Rightarrow \frac{(k+1)(k+2)(2k+3)}{6} + (k+1)^2 \cdot 6$$

$$\Rightarrow \frac{(k+1)(k+2)(2k+3) + 6(k+1)^2}{6}$$

$$\Rightarrow \frac{(k+1)(k+2)(2k+9)}{6}$$

$$\Rightarrow \text{RHS } P(k+1).$$

Q-3 A die is rolled once. what are the probabilities of the following events?

a. Getting an odd number

b. Getting at least a value 2

c. Getting at most a value 2

d. Getting at least 7

Ams- In a standard six-sided die, the possible outcomes are the numbers 1 through 6.

Let's calculate the probabilities for the given events:

** a. Getting an odd number. **

- odd number on a die: 1, 3, 5

- Number of favourable outcomes =
3

Total possible outcomes = 6

$P(\text{odd number}) = \frac{\text{Number of favorable outcomes}}{\text{Total possible outcomes}}$

$$= \frac{3}{6} = \frac{1}{2}$$

** b. Getting at least a value 2. **

- Number at least 2: 2, 3, 4, 5, 6

- Number of favourable outcome = 5

- Total possible outcomes = 6

$P(\text{At least 2}) = \frac{\text{Number of favourable outcomes}}{\text{Total possible outcomes}}$

$$= \frac{5}{6} = \frac{5}{6}$$

c) ** Getting at most a value 2. **

- Number at most 2: 1, 2

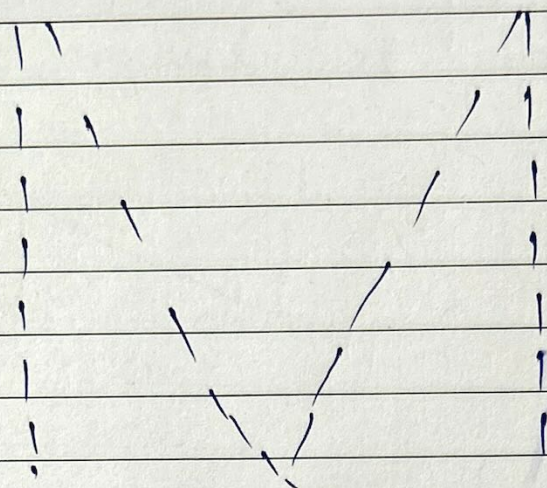
Q-4 Draw a hypercube graph Q_3 (also called the cubical hypercube). Check whether the hypercube Q_3 is Hamiltonian.

=) The hypercube graph Q_3 , also known as the cubical hypercube, is a graph that represents a three dimensional hypercube. In general Q_n represents an n -dimensional hypercube graph.

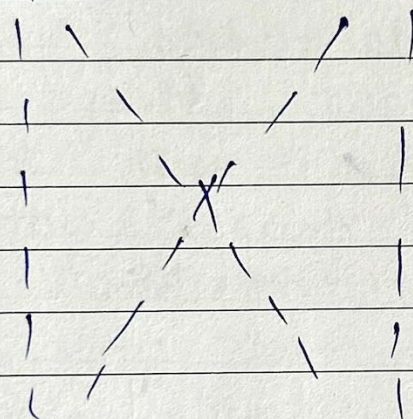
Hypercube graph Q_3 :

In Q_3 , each vertex represents a binary string of length 3. Two vertices are connected by an edge if and only if their binary strings differ in exactly one bit. Let's draw the graph.

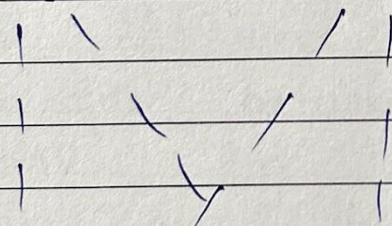
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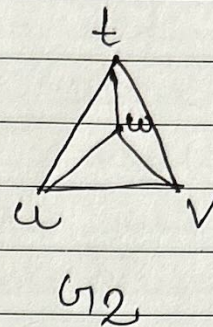
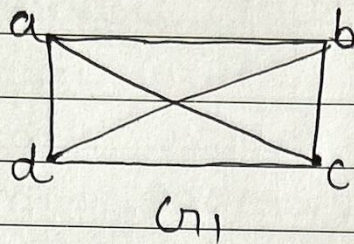


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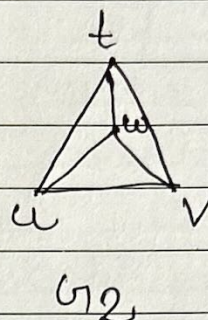
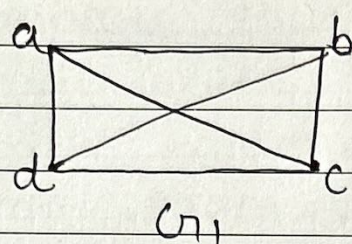
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Q-5 what is isomorphism? Find, if the following graphs G_1 and G_2 are isomorphic or not. Explain how you arrived at your answer.



Ans- Isomorphism in graph theory refers to a structural similarity between two graphs. Two graphs G_1 and G_2 are said to be isomorphic if there exists a bijection (one-to-one and onto) function f that maps the vertices of G_1 to the vertices of G_2 , such that for every pair of vertices u and v in G_1 , there is an edge between them if and only if there is an edge between $f(u)$ and $f(v)$ in G_2 . In simple terms, two graphs are isomorphic if they

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To determine whether the graphs G_1 and G_2 in the image are isomorphic, we can follow these steps:

1.

compare the number of vertices and edges. Both graphs have the same number of vertices (5) and edges (7). This is a necessary condition for isomorphism, but not sufficient.

2.

compares the degrees of the vertices. The degree of a vertex is the number of edges connected to it. In G_1 , the degrees of the vertices are 2, 2, 3, 2 and 2. In G_2 , the degrees of the vertices are 2, 2, 3, 2 and 2. Again, we see that both graphs have the same degree sequence, but this is not enough to prove isomorphism.

3.

Try to find an isomorphism.

Q-6 what is a finite automata? why is it needed? How is a finite automata represented? Also explain the term regular expression with the help of an example.

Ams- ** Finite Automaton (Finite state Machine): **

Key applications and reasons for using finite automata include:

1. ** Language Recognition: ** Finite automata are used to recognize whether a given string belongs to a specified language.

2. ** Lexical Analysis ** → In compiler design, finite automata are employed for lexical analysis to recognize and tokenize input source code.

3. Finite automata are used in pattern recognition.

Q-7 Differentiate between

Ans. ^{**} Deterministic Finite Automaton (DFA). ^{**}

A Deterministic Finite Automaton (DFA) is a finite-state machine that recognizes or accepts a set of strings, where each input string causes the machine to go through a sequence of states from the initial state to a final state.

The key characteristics of a DFA include:

1. ^{**} Deterministic Transitions: ^{**}

For each state and input symbol, there is exactly one transition to a next state.

In other words, the transition is uniquely determined by the current state.

Q-8 Describe the divide - and - conquer approach to solve recurrences? Explain how this approach can be used to apply binary search in a sorted list.

Ams- ** Divide - and - Conquer Approach to Solve Recurrences: **

The divide - and - conquer approach is a problem - solving technique that involves breaking down a problem - solving them independently, and combining their solutions to solve the original problem. This technique is often used to design algorithms and solve recurrences.

1. Divide **:

Break the problem into smaller subproblems that are similar to the original problem.

Q-9 what is proposition? Explain with the help of an example. Explain Disjunction and conjunction with the help of truth table for each.

Ans- ** Proposition. **

A proposition is a declarative sentence or statement that is either true or false, but not both.

** Examples: **

1. ** Simple Propositions: **

• p : "The Sky is blue."

• q : "it is raining."

• r : " $2 + 2 = 5$."

2. ** compound Propositions: **

• $p \wedge q$: "The sky is blue and it is raining."

Q-10 Prove the following theorem by direct proof method: "The square of an even integer is an even integer."

Ans- ** Theorem: **

"The square of an even integer is an even integer."

Proof:

Let m be an even integer. By definition, an even integer can be written as $m = 2k$ for some integer k . We want to show that m^2 is also an even integer.

Now, square the expression $m = 2k$

$$m^2 = (2k)^2$$

Simplify the expression:

$$m^2 = 4k^2$$

Q-11 Given the Boolean expression $(a' \vee (bnc')) \vee (bvd')$, draw the corresponding circuit, where a, b, c , and d are the inputs to the circuit.

Ans- To draw the corresponding circuit for the given Boolean expression $(a' \vee (bnc')) \vee (bvd')$, we can break it down into smaller components based on the operations involved.

Let's define intermediate variables for the subexpressions to make the construction clearer:

1. Let $x = bnc'$ (AND operation)
2. Let $y = a' \vee x$ (OR operation)
3. Let $z = bvd'$ (OR operation)
4. The final result is $y \vee z$ (OR operation)

Now, let's draw the circuit using these intermediate variables.

Q-12 Define the terms Domain, Co-domain and Range in the context of a function. Also, find the domain, co-domain and range for a function A to B , where $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16, 25\}$.

Ans. In the context of a function, the terms "domain", "co-domain" and "range" are used to describe various aspects of the functions input and output.

1. ** Domain: **

The domain is the set of all possible input values.

◦ For this function, $\text{dom}(f) = A = \{1, 2, 3, 4\}$

◦ Co-domain ($\text{codom}(f)$):

◦ The co-domain is the set of all possible output values.

Q-13 A committee consisting of 2 male and 2 female workers is to be constituted from 8 male and 9 female workers. In how many distinct ways can this be done?

Ans- To determine the number of distinct ways to form a committee consisting of 2 males and 2 females from a group of 8 males and 9 females, you can use combinations.

The formula for combinations is given by:

$$C(n, r) = \frac{n!}{r!(n-r)!}$$

where $n!$ represents the factorial of n , which is the product of all positive integers up to n .

Q-14. In a tennis tournament, each entrant plays a match in the first round. Next, all winners from the first round a second - round match. winners continue to move on to the next round, until finally only player is left as the tournament winner.

=> In the first round, each player participates in a match, and half of them are eliminated, therefore, after the first round, $\frac{n}{2}$ players remain.

In subsequent rounds, the same process is repeated until only one player is left. The recurrence relation for $T(n)$ can be expressed as follows:

$$T(n) = 1 + 1\left(\frac{n}{2}\right)$$

Q-15 Show, using the pigeonhole principle, that in any group of 30 people, 8 people can always be found who were born on the same day of the week.

Ans →

The pigeonhole principle states that if you distribute n items into m containers and $n > m$, then at least one container must contain more than one item.

There are 7 days of the week, so we have 7 "containers". Each person can be considered as a "pigeon" to be placed into one of the "pigeon holes" (days of the week).

Now let's consider the worst-case scenario, where each of the 30 people is born on a different day of the week. In this case.

Q-16 Define the following in the context of graph & with the help of an example:

- (a) complete graph
- (b) Degree of a vertex
- (c) cycle
- (d) path
- (e) circuit

Ams-

In the context of graph theory, the following terms are defined:

(a) ****complete graph****

A complete graph is a graph in which there is a unique edge between every pair of distinct vertices.

◦ In a complete graph with n vertices, there are $\frac{n(n-1)}{2}$ edges.

Example: The graph K_3 is a complete graph with 3 vertices.

Q-17

what is a bipartite graph? Explain with the help of an example. Give one or two applications of bipartite graphs.

Ans-

A bipartite graph is a type of graph whose vertices can be divided into two disjoint sets, such that every edge connects a vertex from one set to a vertex in the other set. In other words, there are no edges within the sets.

This division of vertices into two sets creates a bipartition of the graph.

Formally, a graph G is bipartite if its vertex set V can be partitioned into two sets U and $V - U$ (where $V - U$ denotes the complement of U), such that every edge in G connects a vertex.

Q-18

How Hamiltonian graphs differ from the Eulerian graphs? Give Dirac's and Ore's criterion for the Hamiltonian graphs.

Ans-

** Hamiltonian Graph vs Eulerian Graphs: **

** Hamiltonian Graphs: **

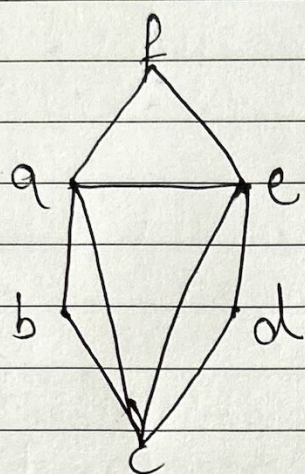
A Hamiltonian graph is a graph that contains a Hamiltonian cycle, which is a cycle that visits every vertex exactly once and returns to the starting vertex.

A Hamiltonian path is a path that visits every vertex exactly once but does not necessarily return to the starting vertex.

- Determining whether a graph is Hamiltonian is a known NP-complete problem.

Q-19

Differentiate between Eulerian graph and Eulerian circuit. Find the Eulerian circuit in the graph given below.



Ans

Here are the steps for finding an Eulerian circuit in a graph:

1. Start at any vertex in the graph.

2. Follow a path that uses edges that have not been used, then you have found an Eulerian circuit.

3. If you are not at the starting vertex.

Q-20 write short notes on the following

(a) Travelling Salesman Problem

(b) vertex coloring

(c) edge coloring

(d) Planar graphs

(e) Pascal's Formula

Ans → ** Traveling Salesman Problem (TSP): **

- The traveling Salesman Problem is a classic optimization problem in graph theory and operations research.

- In TSP, the task is to find the shortest possible tour that visits a set of cities and returns to the starting city, visiting each city exactly once.